Scalability & Future Plan

Date	25/10/2026
Team ID	Tanishka Jain
Project Name	**Electric Vehicle Market Performance & Insights**
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Static Local Data Extracts: Raw EV data is imported from manual spreadsheets (.csv/.xlsx) stored locally on your desktop.	New global vehicle launches, market sales, and battery specs must be manually updated and re-extracted into the workbook.	High
2	**Desktop-Bound Workbook Hosting:** The project lives primarily inside a localized .twbx workbook format.	Reviewers must open the physical file in Tableau Desktop or Tableau Reader, restricting frictionless stakeholder collaboration.	Medium
3	**Row-Level Calculation Performance:** Heavy Level of Detail (LOD) formulas run on the client side at runtime.	As millions of row-level records are added to the dataset, dashboard rendering speeds will degrade during filter actions.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Local file sharing or single-user view via Tableau Public.	Publish workbook to **Tableau Cloud** or **Tableau Server** using Row-Level Security (RLS) to manage hundreds of users simultaneously.
2	Data Storage	Bound by local desktop hard drive space using static flat files.	Migrate backend data assets to a cloud database (e.g., **Snowflake** or **Google BigQuery**) and connect directly via Tableau.
3	Performance	Desktop RAM processing client-side dashboard queries.	Implement a **Tableau Hyper Extract Schedule** or pre-aggregate heavy calculations upstream in the database before Tableau reads it.
4	Security	Open .twbx file where anyone can unpack and view the raw dataset.	Use **Tableau Cloud User Permissions** to encrypt connections and restrict access to sensitive commercial manufacturer data.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Automated Pipelines & Cloud Hosting	1 Month	Transition data to a cloud warehouse with automated weekly schedules, freeing up the developer from manual file updates.
Phase 3	advanced Predictive Analysis	3 Month	Integrate Tableau with **Python/R (TabPy)** to build battery life decay projections and forecasting charts for market sales.
Phase 4	Real-Time Infrastructure Mapping	6 Month	Implement live API data feeds connecting regional charging network station metrics to map out real-time infrastructure growth gaps.